

From: Doug Langworthy <dlangworthy@Cadillac-MI.net>
Sent: 4/3/2020 10:40:39 AM
To: "Sneller, Cindy (EGLE)" <SNELLERC@michigan.gov>
Subject: RE: Documentation needed to continue the RMP Mod Review
Attachments: Fall 19 hauling data.pdf, Fall 19 Path.pdf, TVS Sludge.xlsx, Formula Example.xls

Cindy,
I think I have everything you requested. Let me know if there is anything else.
Thanks Doug

Douglas Langworthy

Assistant Water Resources Supervisor
City of Cadillac
231-775-2841

From: Sneller, Cindy (EGLE) <SNELLERC@michigan.gov>
Sent: Friday, April 3, 2020 7:41 AM
To: Doug Langworthy <dlangworthy@Cadillac-MI.net>
Cc: Person, Michael (EGLE) <PERSONM@michigan.gov>
Subject: Documentation needed to continue the RMP Mod Review

Hi Doug – Mike Person and I discussed the proper documentation for review of the RMP Mod. [The following would be necessary in order to continue the RMP Mod Review.](#)

A minimum of *two years of data/documentation* regarding: Pathogen results (Rule 323.2414(2)(f) Alternative 4) [and](#) Vector Attraction Reduction (VAR) – 323.2415(4)(a) Reduction of Volatile Solids by 38%.

- [Data/documentation](#) showing how the formula is used with the pre- and post- sludge data, to achieve compliance.
- This may be:
 - Copies of the lab bench sheets (pre- and post- sludge data), along with the Excel spreadsheet calculations showing compliance with the 38% reduction. **Or,**
 - The plant's Excel spreadsheet (if all lab work is electronic), where we can see the data (pre- and post- sludge data), and access the formula used to calculate the reduction and achieve compliance. **Or,**
 - However the WWTP documents the data and the calculation, for compliance to be achieved.

Provide also a written statement of the formula used to calculate how the Volatile solids reduction is met.

- [What is that formula?](#)

Currently our office has: an excel spreadsheet of the **pre**-treated (raw) sludge data for June, 2019 and the **post**-treated sludge data during the land application of June 2019 (which you provided), and We have access to the pathogen results in the last two years via the fiscal year biosolids annual reports; According to the Biosolids Annual Reports, pathogen samples were collected twice each year.

- 2018 Pathogens were collected: March and September;

- 2019 Pathogens were collected: April and Fall (month?);

There is minimal data/documentation for VAR in these biosolids annual reports.

- The post-treated sludge data (during the hauling week);
- no pre-treated sludge data;

At this point, our office would like the following VAR data (pre and post Volatile Sludge results) for these time periods:

1. **Year 2018** – hauling occurred: May and October, 2018;
 - Please provide VAR data/documentation for **May and October, 2018;**
2. **Year 2019** – hauling occurred: June and Fall 2019;
 - Our office has the June 2019 data/documentation;
 - Please provide VAR data/documentation for the **Fall (month?) haul of 2019;**

Once we have that data, I will be able to continue with the review and hopefully have it completed soon after that.

If you have any questions regarding the above, feel free to contact me. I'm available today until approx.. 2:30 pm, then back on Monday.

Cindy

Cindy A. Sneller, Senior EQA
EGLE/WRD/Biosolids - Covering the west side of the Lower Peninsula
snellerc@michigan.gov
616-401-2471

Grand Rapids District Office
State Office Building
350 Ottawa Ave., NW
Grand Rapids, MI 49503
www.michigan.gov/egle

ATTACHMENT NAME:

Fall 19 hauling data.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

BIO SOLIDS LAB DATA

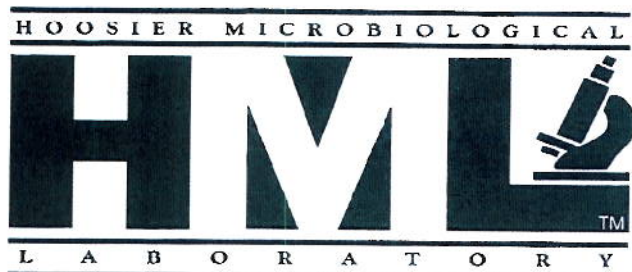
[illegible]

ATTACHMENT NAME:

Fall 19 Path.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image



Testing • Research • Consulting

Multiple Analysis Report

Sample: 382164

November 13, 2019

Ms. Cindy Tomaszewski
City of Cadillac
200 N. Lake St.
Cadillac, MI 49601

RE: PWS ID#: Unavailable
City of Cadillac
Digester #4
1121 Plett Rd.
Cadillac, MI 49601

Dear Ms. Tomaszewski:

The following are the result(s) of the test(s) performed on the sample(s) received at HML, Inc. at 10:20 AM, 09/26/2019, and collected at 1:15 PM, 09/25/2019:

<u>TEST - METHOD</u>	<u>RESULT</u>	<u>MDL*</u>	<u>Date Complete</u>
Fecal Coliform-EPA 1680	<1 MPN/g dry	1 MPN/g dry	09/30/2019
Enteric Virus-EPA 625/R-92/013	<1 pfu/4g dry	1 pfu/4g dry	11/12/2019
Solids, total-SM 2540G	2.30%	0.005%	10/01/2019
Salmonella-EPA 1682	<3 MPN/4 g dry	3 MPN/4 g dry	10/01/2019
Helminth Ova-EPA/625/R-92/013	<1 ovum/4g dry	1 ovum/4 g dry	11/07/2019

*Minimum Detection Level

This testing was completed by T.K. and K.L. Please feel free to contact us if we can be of further service to you.

Sincerely,

Donald A. Hendrickson, Ph.D.
President - Microbiologist
Chemistry Lab #C-18-01
Microbiological Lab #M-18-03

DAH/skp

ATTACHMENT NAME:

TVS Sludge.xlsx

ATTACHMENT TYPE:

Microsoft Office 12+, Office Open XML (OOXML/OpenXML): spreadsheet (XLSX)

5/18

DATE

RAW SLUDGE	V. RAW SLUDGE	RAW SLDGE	DIG
T.S.	T.V.S.	PUMPED	TEMP
2.93	77.19	14400.00	97.00
3.11	78.33	14400.00	97.00
3.19	74.39	14400.00	97.00
3.44	78.79	14400.00	96.00
		14400.00	97.00
		14400.00	97.00
3.03	75.81	14400.00	96.00
3.31	79.31	14400.00	95.00
3.12	79.69	14400.00	96.00
3.18	72.86	14400.00	95.00
2.99	75.44	14400.00	95.00
		14400.00	95.00
		14400.00	95.00
3.02	77.19	14400.00	95.00
3.20	75.81	14400.00	95.00
3.24	75.81	14400.00	95.00
3.09	75.41	14400.00	95.00
3.01	76.19	14400.00	95.00
		14400.00	97.00
		14400.00	96.00
2.70	75.38	14400.00	97.00
3.22	75.81	14400.00	97.00
2.70	75.86	17600.00	97.00
3.00	74.19	14400.00	98.00
3.26	72.13	14400.00	98.00
		14400.00	94.00
		14400.00	94.00
		14400.00	98.00
2.97	76.47	14400.00	98.00
3.10	71.67	14400.00	97.00
3.07	73.68	14400.00	95.00
67.88	1667.41	449600.00	2979.00
3.09	75.79	14503.23	96.10

TRANS	SUPER	DECANT
1 - 2	#2	# 3 or # 4
56730.00	10065.00	2400.00
	3695.00	4200.00
	11230.00	
55205.00	1220.00	17400.00
		9600.00
	7625.00	
	5490.00	
51850.00	7930.00	
	9455.00	
	11285.00	
39955.00	15250.00	
	8845.00	
	8540.00	
63440.00	6100.00	
	7930.00	
	6710.00	
	8845.00	
42395.00	5490.00	
	2440.00	
	4880.00	
41175.00	5490.00	
	5490.00	
	3695.00	
34465.00	9760.00	
30500.00	9455.00	
	6710.00	
	3660.00	
28060.00	6100.00	
	9760.00	
19825.00	8845.00	
	7930.00	
463600.00	219920.00	33600.00
42145.45	7330.67	8400.00

10/18
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RAW SLUDGE T.S.
2.28
2.17
2.20
2.14
2.25
2.19
2.16
2.14
2.10
1.80
2.42
2.21
2.33
2.43
2.28
2.28
2.25
2.29
2.35
2.36
2.30
2.41
2.38
51.72
2.25

V. RAW SLUDGE T.V.S.	RAW SLUDGE PUMPED	DIG TEMP	TRANS 1 - 2	SUPER #2	DECANT # 3 or # 4
71.11	14400.00	98.00	19825.00	8845.00	
73.33	14400.00	98.00		15555.00	
74.74	14400.00	98.00	50630.00	7015.00	3600.00
73.17	14400.00	98.00		8235.00	
76.09	14400.00	98.00	29280.00	14640.00	
	14400.00	98.00		10980.00	
	14400.00	98.00		10980.00	
71.74	14400.00	98.00	47275.00	3965.00	2400.00
72.09	14400.00	98.00		9760.00	
64.10	14400.00	98.00		6405.00	4200.00
77.27	14400.00	98.00		14335.00	
75.00	14400.00	98.00	39650.00	4880.00	5400.00
	14400.00	99.00		12200.00	
	14400.00	98.00		9760.00	
68.75	14400.00	98.00	23180.00	3355.00	
71.11	14400.00	98.00		10065.00	
74.07	14400.00	99.00	17995.00	5795.00	
70.59	14400.00	99.00	51545.00	7930.00	
70.83	14400.00	98.00		6710.00	
	14400.00	98.00		12810.00	
	14400.00	98.00		12810.00	
72.92	14400.00	98.00	40260.00	5490.00	
73.33	14400.00	98.00		6100.00	
71.15	14400.00	98.00		11590.00	
73.47	14400.00	98.00		5490.00	
71.43	14400.00	98.00	53375.00	2745.00	
	14400.00	98.00		8540.00	
	14400.00	98.00		9760.00	
71.74	14400.00	98.00		9150.00	
69.39	14400.00	98.00	28060.00	4575.00	
72.34	14400.00	98.00		4270.00	
1659.76	446400.00	3041.00	401075.00	264740.00	15600.00
72.16	14400.00	98.10	36461.36	8540.00	3900.00

3/19

DATE	RAW SLUDGE	V. RAW SLUDGE
	T.S.	T.V.S.
1	3.26	68.18
2		
3		
4	3.27	80.28
5	3.26	77.27
6	3.22	78.26
7	3.16	77.53
8	3.07	78.69
9		
10		
11	3.31	76.12
12	3.10	77.42
13	2.91	76.67
14	3.09	77.42
15	3.57	73.97
16		
17		
18	3.13	74.58
19	2.74	78.69
20	3.27	78.46
21	3.20	78.74
22	3.21	78.26
23		
24		
25	3.02	78.33
26	3.04	76.19
27	3.15	75.81
28	3.03	76.21
29	2.99	79.03
30		
31		
	65.99	1616.10
AVG	3.14	76.96

4/19

DATE	RAW SLUDGE	V. RAW SLUDGE
	T.S.	T.V.S.
1	2.89	68.85
2	3.19	77.61
3	3.46	73.53
4	3.13	75.41
5	2.38	76.09
6		
7		
8	3.40	78.79
9	3.18	77.97
10	3.22	77.05
11	3.03	87.93
12	3.35	77.33
13		
14		
15	2.94	80.00
16	3.17	79.03
17	3.17	81.67
18	2.96	78.33
19		
20		
21		
22	2.98	80.88
23	2.84	78.08
24	3.10	77.97
25	2.89	80.00
26	3.03	77.97
27		
28		
29	3.41	70.77
30	3.75	72.73
	65.47	1627.99
AVG	3.12	77.52

DATE	RAW SLUDGE	V. RAW SLUDGE	RAW SLUDGE	DIG	TRANS
	T.S.	T.V.S.	PUMPED	TEMP	1 - 2
1			14400.00	98.00	
2	2.19	76.74	14400.00	98.00	
3	2.27	74.47	14400.00	98.00	39040.00
4	2.26	74.42	14400.00	98.00	
5	2.24	74.42	14400.00	98.00	
6	2.36	72.73	14400.00	98.00	39955.00
7			14400.00	98.00	
8			14400.00	98.00	
9	2.28	73.91	14400.00	98.00	
10	2.55	75.00	14400.00	98.00	
11	2.48	76.60	19200.00	98.00	39955.00
12	2.75	69.64	19200.00	96.00	
13	2.37	74.00	14400.00	96.00	53070.00
14			14400.00	97.00	
15			14400.00	97.00	
16	2.65	77.65	14400.00	98.00	
17	2.75	74.51	19200.00	98.00	36905.00
18	2.75	75.44	14400.00	97.00	49410.00
19	2.82	75.00	14400.00	97.00	
20	2.91	77.05	14400.00	98.00	
21			14400.00	98.00	33550.00
22			14400.00	98.00	
23	2.87	78.33	14400.00	97.00	
24			14400.00	97.00	41175.00
25			19200.00	97.00	
26	2.84	75.93	14400.00	97.00	
27	2.76	74.51	14400.00	98.00	58560.00
28			14400.00	97.00	
29			14400.00	97.00	
30	2.91	77.59	14400.00	97.00	
			451200.00	2925.00	391620.00
	49.01	1427.94	15040.00	97.50	43513.33
	2.58	75.15			

11/19

SUPER #2	DECANT # 3 or # 4
3965.00	
5185.00	
9760.00	
10675.00	
10675.00	
10675.00	
	9600.00
13420.00	
10370.00	
6710.00	
6405.00	
	12000.00
4880.00	6000.00
6100.00	6000.00
7320.00	
11590.00	
15250.00	
5185.00	4800.00
10370.00	
15555.00	
13115.00	
12200.00	
14030.00	
11285.00	
7930.00	4800.00
4880.00	
12200.00	
7015.00	
6405.00	
8540.00	
261690.00	43200.00
9346.07	7200.00

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RAW SLUDGE T.S.	V. RAW SLUDGE T.V.S.
2.11	72.09
1.93	78.05
1.93	75.00
1.98	75.00
1.88	74.65
1.98	72.50
1.74	75.68
1.97	75.00
1.69	78.24
1.97	80.00
2.06	78.05
1.86	81.08
1.95	75.00
1.77	77.46
1.96	78.95
2.06	76.74
1.66	81.82
1.83	82.21
34.32	1387.51
1.91	77.08

RAW SLUDGE PUMPED	DIG TEMP	TRANS 1 - 2	SUPER #2	DECANT # 3 or # 4
14400.00	98.00	52155.00	2440.00	7200.00
14400.00	98.00		2135.00	9600.00
14400.00	98.00		6710.00	
16800.00	98.00	41480.00	11590.00	6000.00
16800.00	97.00		5490.00	6000.00
16800.00	97.00			9600.00
19200.00	97.00	10370.00	6710.00	9600.00
19200.00	97.00	58865.00		9600.00
19200.00	98.00		7015.00	10800.00
19200.00	97.00	46055.00	7015.00	6000.00
19200.00	97.00		7930.00	10800.00
14400.00	97.00	55815.00		16200.00
14400.00	97.00		15860.00	4200.00
14400.00	97.00		11590.00	4688.00
14400.00	97.00	57645.00	14640.00	3750.00
14400.00	97.00		5490.00	1875.00
14400.00	97.00		10675.00	
19200.00	97.00	36905.00	19215.00	
19200.00	97.00		13725.00	
19200.00	97.00	33550.00	18910.00	
19200.00	96.00	18605.00	18300.00	
19200.00	95.00	71675.00	1830.00	
19200.00	96.00		4270.00	
19200.00	95.00		9150.00	
19200.00	92.00	45750.00	2440.00	
19200.00	92.00	11285.00	5185.00	
14400.00	92.00	59170.00	11590.00	
19200.00	92.00		8540.00	
19200.00	92.00			
14400.00	92.00	45140.00		
516000.00	2879.00	644465.00	228445.00	115913.00
17200.00	95.97	42964.33	9137.80	7727.53

ATTACHMENT NAME:

Formula Example.xls

ATTACHMENT TYPE:

Microsoft Office 1-11, Binary format: spreadsheet (XLS)

Formula= $\frac{\%IN-\%OUT}{\%IN-(\%IN \times \%OUT)}$

Volatile Solids	
IN = 72.16%	Volatile Solids Reduction = 50.25%
OUT = 56.32%	

Avg Raw= 72.16%
Avg Hauling= 56.32%
Reduction= 50.25%

Oct-18	
DATE	V. RAW SLUDGE T.V.S.
1	71.11
2	73.33
3	74.74
4	73.17
5	76.09
6	
7	
8	71.74
9	72.09
10	64.10
11	77.27
12	75.00
13	
14	
15	68.75
16	71.11
17	74.07
18	70.59
19	70.83
20	
21	
22	72.92
23	73.33
24	71.15
25	73.47
26	71.43
27	
28	
29	71.74
30	69.39
31	72.34
	1659.76
AVG	72.16

Hauling 10/18	
DATE	Digested T.V.S.
10/16/2018	55.81
10/17/2018	54.76
10/18/2018	58.14
10/19/2018	55
10/20/2018	57.89
Composite	54.17